



Year 12 Mathematics Extension 2
2014 HSC ASSESSMENT TASK 1
Term 4 Week 10 2013

Name: _____

Teacher: _____

Setter: VUL Wed 4th December Periods 4 and 5

- Attempt **all** questions.
- All questions are of equal value.
- Marks may be deducted for insufficient, or illegible work.
- Only Board approved calculators may be used
- Total possible mark is **50**
- **Begin each question on a new page.**
- **TIME ALLOWED** : 90 minutes + 2 minutes reading time.

Question 1

Marks

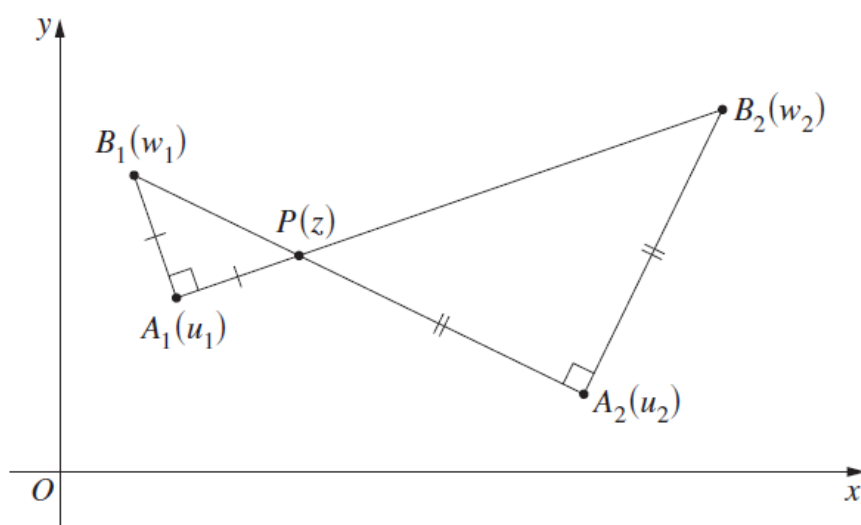
- (a) Let $z = \sqrt{3} - i$ and $w = i - 1$
- (i) Find $z - \bar{w}$ in the form $x + iy$ where x and y are real. 1
- (ii) Find $\frac{z}{w}$ in the form $x + iy$ where x and y are real 2
- (iii) Write z and w in modulus-argument form 2
- (iv) Hence or otherwise show that $\frac{(\sqrt{3} - i)^{30}}{(i - 1)^{50}}$ is purely imaginary 3
- (b) Solve $2z^2 - (3 + i)z + 2 = 0$ 4
- (c) Find the Cartesian equations and give a geometric description of the locus of points z such that:
- (i) $|z + 2| = |z - i|$ 2
- (ii) $|z + 2| = 2|z - i|$ 2
- (d) Sketch the locus of points z so that the inequalities $|z + \bar{z}| \leq 2$ and $\frac{\pi}{4} \leq \arg(z - i) \leq \frac{3\pi}{4}$ hold simultaneously 3
- (e) On an Argand diagram plot the position of a complex number z with $0 < \operatorname{Re}(z) < 1$, $0 < \operatorname{Im}(z) < 1$ and $\frac{\pi}{4} < \arg(z) < \frac{\pi}{2}$. Then plot z^2 on the same Argand diagram. 2
- (f) Prove by mathematical induction that $\sum_{r=1}^n \frac{1}{r^2} \leq 2 - \frac{1}{n}$ for all integers $n \geq 1$ 4
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Question 2 **Start a new page**

Marks

- (a) On the Argand diagram the points A_1 and A_2 correspond to the distinct complex numbers u_1 and u_2 respectively. Let P be a point corresponding to a third complex number z .

Points B_1 and B_2 are positioned so that $\triangle A_1PB_1$ and $\triangle A_2B_2P$, labelled in an anti-clockwise direction, are right-angled and isosceles with right angles at A_1 and A_2 , respectively. The complex numbers w_1 and w_2 correspond to B_1 and B_2 , respectively.



- (i) Explain why $w_1 = u_1 + i(z - u_1)$. 2
- (ii) Find the locus of the midpoint of B_1B_2 as P varies. 2
- (b) If $z = \cos \theta + i \sin \theta$:
- (i) Show that $z^n - \frac{1}{z^n} = 2i \sin n\theta$ 2
- (ii) By expanding $\left(z - \frac{1}{z}\right)^3$ express $\sin^3 \theta$ in terms of $\sin n\theta$ 3
- (c) (i) Given that ω is a complex cube root of unity, show that ω^2 is the other complex root. 2
- (ii) Plot the three cube roots of unity on an Argand diagram 1
- (iii) Show that $1 + \omega + \omega^2 = 0$ 1
- (iv) Simplify $(7 + 9\omega^4 + 7\omega^{-1})^6$ 2

Question 2 continued on the next page

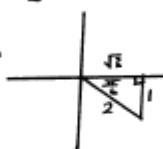
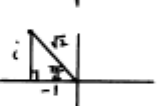
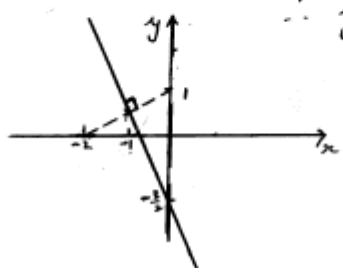
Question 2 Continued

- (d) (i) Solve $z^6 + 1 = 0$ over the set of complex numbers. 3
- (ii) Hence or otherwise express $z^6 + 1$ as product of real quadratic factors. 3
- (e) Let $z_2 = 1 + i$ and, for $n > 2$, let $z_n = z_{n-1} \left(1 + \frac{i}{|z_{n-1}|} \right)$. 4

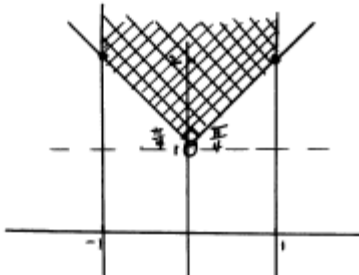
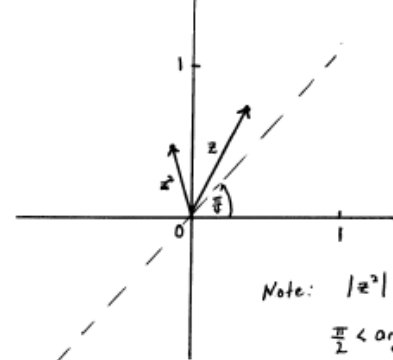
Use mathematical induction to prove that $|z_n| = \sqrt{n}$ for all integers $n \geq 2$.

End of Assessment Task

2014 (2013 Term4) Year 12 Mathematics Extension 2 HSC Task 1 Solutions

Suggested Solution (s)	Comments	Suggested Solution (s)
Question 1 a) i) $z - \bar{w} = \sqrt{3} - i - (-1 - i)$ $= 1 + \sqrt{3}$ ✓ ii) $\frac{z}{w} = \frac{\sqrt{3} - i}{-1 + i} \times \frac{-1 - i}{-1 - i}$ ✓ $= \frac{-\sqrt{3} - \sqrt{3}i - i - 1}{2}$ $= \frac{-\sqrt{3} - 1}{2} - \frac{(\sqrt{3} + 1)}{2}i$ ✓ iii) $z = \sqrt{3} - i$ $= 2 \operatorname{cis}\left(-\frac{\pi}{6}\right)$ ✓  $w = -1 + i$ $= \sqrt{2} \operatorname{cis}\left(\frac{3\pi}{4}\right)$ ✓  iv) $\frac{(\sqrt{3} - i)^{30}}{(i - 1)^{50}} = \frac{[2 \operatorname{cis}(-\frac{\pi}{6})]^{30}}{[\sqrt{2} \operatorname{cis}(\frac{3\pi}{4})]^{50}}$ $= \frac{2^{30} \operatorname{cis}(-5\pi)}{2^{25} \operatorname{cis}(\frac{15\pi}{2})}$ by DMT ✓ $= 2^5 \operatorname{cis}(-5\pi - \frac{15\pi}{2})$ $= 2^5 \operatorname{cis}(-\frac{85\pi}{2})$ ✓ $= 32 \operatorname{cis}(-\frac{\pi}{2})$ $= -32i$ ✓ $\therefore$ Purely Imaginary.		(b) $2z^2 - (3+i)z + 2 = 0$ $z = \frac{3+i \pm \sqrt{(3+i)^2 - 4(2)(2)}}{2(2)}$ $= \frac{3+i \pm \sqrt{-8+6i}}{4}$ ✓ Let $(x+iy)^2 = -8+6i$ $\therefore x^2 - y^2 + 2xyi = -8+6i$ $\therefore x^2 - y^2 = -8 \quad xy = 3$ ✓ $x = \pm 1, y = \pm 3$ by inspection $\therefore z = \frac{3+i+1+3i}{4}$ or $\frac{3+i-1-3i}{4}$ $= 1+i$ or $\frac{1}{2} - \frac{1}{2}i$ ✓ c) i) $z+2 = z-i$ is perpendicular bisector of $(-2, 0)$ and $(0, 1)$ ✓ $\therefore M = -\frac{-2-0}{0-(1)} = 2$ $M\left(\frac{-2+0}{2}, \frac{0+1}{2}\right)$ $= M\left(-1, \frac{1}{2}\right)$ $\therefore y - \frac{1}{2} = 2(x+1)$ $y = -2x + \frac{5}{2}$ ✓ 

2014 (2013 Term4) Year 12 Mathematics Extension 2 HSC Task 1 Solutions

Suggested Solution (s)	Comments	Suggested Solution (s)
ii) $z+2 = 2 z-i$ let $z = x+iy$ $\therefore x+2+iy = 2 x+i(y-1)$ $\sqrt{(x+2)^2 + y^2} = 2\sqrt{x^2 + (y-1)^2}$ $x^2 + 4x + 4 + y^2 = 4(x^2 + y^2 - 2y + 1)$ $x^2 + 4x + 4 + y^2 = 4x^2 + 4y^2 - 8y + 4$ $3x^2 + 3y^2 - 4x - 8y = 0$ $x^2 - \frac{4}{3}x + \frac{16}{9} + y^2 - \frac{8}{3}y + \frac{16}{9} = \frac{20}{9}$ $(x - \frac{2}{3})^2 + (y - \frac{4}{3})^2 = \frac{20}{9}$ $\therefore$ Circle Centre $(\frac{2}{3}, \frac{4}{3})$ radius $\frac{2\sqrt{5}}{3}$ ✓ d) let $z = x+iy$ $\therefore x+iy + x-iy \leq 2$ $ 2x \leq 2$ $-2 \leq 2x \leq 2$ $-1 \leq x \leq 1$ ✓  ✓✓ (e)  Note: $z^2 < z$ ✓ $\frac{\pi}{2} < \arg(z) < \pi$ ✓		f) When $n=1$, $\sum_{r=1}^n \frac{1}{r^2} - (2 - \frac{1}{n}) = \frac{1}{1^2} - (2 - \frac{1}{1})$ $= 1 - 1$ $= 0 \leq 0$ $\therefore \sum_{r=1}^n \frac{1}{r^2} \leq 2 - \frac{1}{n}$ for $n=1$ ✓ Assume the statement is true for $n=k$, some fixed positive integer, i.e. $\sum_{r=1}^k \frac{1}{r^2} \leq 2 - \frac{1}{k}$ When $n=k+1$, $\sum_{r=1}^n \frac{1}{r^2} - (2 - \frac{1}{n}) = \sum_{r=1}^k \frac{1}{r^2} + \frac{1}{(k+1)^2} - (2 - \frac{1}{k+1})$ $\leq 2 - \frac{1}{k} + \frac{1}{(k+1)^2} - 2 + \frac{1}{k+1}$ $= \frac{1}{k+1} - \frac{1}{k} + \frac{1}{(k+1)^2}$ by assumption $= \frac{k(k+1) - (k+1)^2 + k}{k(k+1)^2}$ $= \frac{k^2 + k - k^2 - 2k - 1 + k}{k(k+1)^2}$ $= \frac{-1}{k(k+1)^2} \leq 0 \text{ as } k > 1 \text{ and } (k+1)^2 > 0$ $\therefore$ If statement true for $n=k$, It has been proved true for $n=k+1$, Since the statement true for $n=1$, then true for $n=2, 3, 4, \dots$ $\therefore \sum_{r=1}^n \frac{1}{r^2} \leq 2 - \frac{1}{n}$ for $n \geq 1$

Question 2

$$\begin{aligned} \text{a) i) } w_1 &= \vec{OB}_1 \\ &= \vec{OA}_1 + \vec{A_1B_1} \quad \checkmark \quad \text{but } \vec{A_1B_1} = \vec{A_1P} \times i \\ &= u_1 + i(z - u_1) \quad = [\vec{OP} - \vec{OA_1}] \times i \quad \checkmark \\ &= (z - u_1)i \end{aligned}$$

$$\begin{aligned} \text{ii) Similarly } u_2 &= \vec{OB}_2 \\ &= \vec{OA_2} + \vec{A_2B_2} \quad \vec{A_2B_2} = \vec{A_2P} \times -i \\ &= u_2 + i(u_2 - z) \quad \checkmark \quad = [\vec{OP} - \vec{OA_2}] \times -i \\ &= [z - u_2] \times -i \\ &= (u_2 - z)i \end{aligned}$$

Now midpoint of B_1B_2

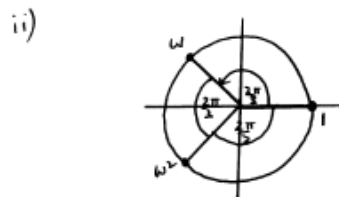
$$\begin{aligned} &= \frac{1}{2} (\vec{OB}_1 + \vec{OB}_2) \\ &= \frac{1}{2} [u_1 + i(z - u_1) + u_2 + i(u_2 - z)] \\ &= \frac{1}{2} (u_1 + u_2 - (u_1 - u_2)i) \quad \checkmark \end{aligned}$$

$$\begin{aligned} \text{b) i) } z^n &= (\cos \theta + i \sin \theta)^n \\ &= \cos n\theta + i \sin n\theta \quad \text{--- ①} \quad \checkmark \\ z^{-n} &= \cos(-n\theta) + i \sin(-n\theta) \quad \checkmark \\ &= \cos n\theta - i \sin n\theta \quad \text{--- ② as } \cos(-A) = \cos A \\ &\quad \sin(-A) = -\sin A \\ \text{So ① + ②: } &\checkmark \quad z^n + z^{-n} = 2 \cos n\theta \\ \therefore z^n - z^{-n} &= 2i \sin n\theta \end{aligned}$$

$$\begin{aligned} \text{ii) } \left(z - \frac{1}{z}\right)^3 &= z^3 - 3z^{\frac{1}{2}} + 3z \cdot \frac{1}{z} - \frac{1}{z^3} \quad \checkmark \\ &= z^3 - \frac{1}{z^3} - 3\left(z - \frac{1}{z}\right) \\ \text{When } n=1 & \quad (2i \sin \theta)^3 = z^3 - \frac{1}{z^3} - 3\left(z - \frac{1}{z}\right) = 2i \sin 3\theta - 3(2i \sin \theta) \quad \checkmark \\ \therefore -8i \sin^3 \theta &= 2i \sin 3\theta - 6i \sin \theta \\ \therefore \sin^3 \theta &= \frac{1}{4} \sin 3\theta - \frac{3}{4} \sin \theta \quad \checkmark \end{aligned}$$

c) i) The three cube roots of unity, z_0, z_1, z_2 are equally spaced $\frac{2\pi}{3}$ radians apart on unit circle with $z_0 = 1$ $\checkmark$

$$\begin{aligned} \therefore z_1 &= \text{cis } \frac{2\pi}{3} = w \\ \text{and } z_2 &= \text{cis } \frac{4\pi}{3} = (\text{cis } \frac{2\pi}{3})^2 \quad \checkmark \\ &= w^2 \end{aligned}$$



iii) If w is a complex ^{cube} root of unity

$$\begin{aligned} \text{then } w^3 &= 1 \\ w^3 - 1 &= 0 \quad \checkmark \\ (w-1)(w^2 + w + 1) &= 0 \\ \therefore w^2 + w + 1 &= 0 \quad w-1 \neq 0 \\ &\quad w \neq 1 \end{aligned}$$

$$\begin{aligned} \text{iv) } (7 + 9w^2 + 7w^{-1})^6 &= \left(7 + 9w + \frac{7}{w}\right)^6 \quad w^2 = 1 \\ &= \left(\frac{7w + 9w^2 + 7}{w}\right)^6 \quad \therefore w^2 = w \\ &= (7(1 + w + w^2) + 2w^2)^6 \quad w^2 = 1 \\ &= (2w^2)^6 \quad w^2 = 1 \\ &= 2^6 \cdot w^{12} \quad w^6 = 1 \\ &= 64 \quad \checkmark \end{aligned}$$

2014 (2013 Term4) Year 12 Mathematics Extension 2 HSC Task 1 Solutions

d) i) If $z^6 + 1 = 0$

$$z^6 = -1 = \text{cis}(\pi + 2n\pi)$$

$$\therefore z_n = \text{cis}\left(\frac{\pi}{6} + n \cdot \frac{\pi}{3}\right) \quad n = 0, 1, 2, \dots, 5 \quad \checkmark$$

$$\therefore z_0 = \text{cis}\frac{\pi}{6} = \frac{\sqrt{3}}{2} + \frac{i}{2}$$

$$z_1 = \text{cis}\frac{\pi}{2} = i$$

$$z_2 = \text{cis}\frac{5\pi}{6} = -\frac{\sqrt{3}}{2} + \frac{i}{2}$$

$$z_3 = \text{cis}\frac{7\pi}{6} = -\frac{\sqrt{3}}{2} - \frac{i}{2}$$

$$z_4 = \text{cis}\frac{3\pi}{2} = -i$$

$$z_5 = \text{cis}\frac{11\pi}{6} = \frac{\sqrt{3}}{2} - \frac{i}{2}$$

ii) Let $z^{k+1} = (z - z_0)(z - z_1)(z - z_2)(z - z_3)(z - z_4)(z - z_5)$ $\checkmark$
 $= (z - \text{cis}\frac{\pi}{6})(z - \text{cis}\frac{\pi}{2})(z - \text{cis}\frac{5\pi}{6})(z - \text{cis}\frac{7\pi}{6})(z - \text{cis}\frac{3\pi}{2})(z - \text{cis}\frac{11\pi}{6})$
 $= [z^2 - (z_0 + z_5)z + z_0 z_5][z^2 - (z_1 + z_4)z + z_1 z_4][z^2 - (z_2 + z_3)z + z_2 z_3]$
 $= (z^2 - \sqrt{3}z + 1)(z^2 + 1)(z^2 + \sqrt{3}z + 1) \quad \checkmark$

e) When $n = 2$,

$$\text{L.H.S} = |z_n|$$

$$= |z_2| \text{ with } z_2 = 1 + i$$

$$= \sqrt{2}$$

$$\text{R.H.S} = \sqrt{n}$$

$$= \sqrt{2}$$

$$\therefore |z_n| = \sqrt{n} \text{ for } n = 2 \quad \checkmark$$

Assume the statement is true for $n = k$ some fixed positive integer

$$\text{i.e. } |z_k| = \sqrt{k}$$

When $n = k+1$ $|z_n| = |z_{k+1}|$

$$= |z_{k+1}| \left(1 + \frac{i}{|z_{k+1}|}\right) \quad \checkmark$$

$$= |z_k| \cdot \left|1 + \frac{i}{|z_k|}\right| \quad \checkmark$$

$$= \sqrt{k} \left|1 + \frac{i}{\sqrt{k}}\right| \text{ by assumption}$$

$$= \sqrt{k} \times \sqrt{1^2 + \left(\frac{1}{\sqrt{k}}\right)^2} \quad \checkmark$$

$$= \sqrt{k} \times \sqrt{1 + \frac{1}{k}}$$

$$= \sqrt{k\left(1 + \frac{1}{k}\right)}$$

$$= \sqrt{k+1}$$

$$= \sqrt{n} \text{ when } n = k+1$$

etc (M2 conclusion).